

Undirected Graph

Graph Theory Explorer — Study Notes

Definition

An undirected graph is a graph in which edges have no direction — each edge represents a two-way (symmetric) connection between two vertices. If an edge connects vertex A to vertex B, you can traverse it from A to B or from B to A equally.

Key Properties

- Edges are represented as unordered pairs: edge $\{A, B\}$ is identical to edge $\{B, A\}$.
- The degree of a vertex equals the number of edges incident to it.
- The Handshaking Lemma applies: the sum of all vertex degrees equals twice the number of edges ($2|E|$).
- Adjacency matrices for undirected graphs are always symmetric.
- Common examples include social networks (mutual friendships) and road maps with two-way streets.

Example

Let $V = \{A, B, C, D\}$ and $E = \{\{A,B\}, \{B,C\}, \{C,D\}, \{D,A\}\}$. This forms a 4-cycle where you can move freely in either direction along each edge — from A to B or B to A, and so on around the cycle.

Why It Matters

Undirected graphs model relationships that are inherently mutual, such as friendships, collaboration networks, or physical connections like cables and roads without one-way restrictions. They contrast with directed graphs (digraphs), where edges represent one-way relationships like 'follows' on social media or one-way streets.